

Technical Document #776: Data Engineering - Comprehensive Overview

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Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Data warehouses store structured data optimized for analytical queries.
- Schema evolution handles changes in data structure over time.
- Data lakes store raw data in its native format.